Midterm Mock Exam: Worked Solutions

Hospital readmission scenario · miss = 9,000 · unnecessary call = 1,000 · 100 points · 100 minutes. Do the mock first under time pressure, then grade yourself here. These are model answers written to the course's high-score standard: a decision, a number, a course concept and a limit on what you claim.

Time plan for the real exam (100 min): Part A 10 min · Part B 35 min · Part C 30 min · Part D 20 min · 5 min to check arithmetic. Points per minute are roughly equal across parts, so never get stuck on one item.

Part A · Concept checks (12 p)

#	Answer	Why (the concept being tested)
1	**B**	Leakage. "Follow-up calls made after discharge" happens *after* the prediction time. 88% → 99% is the leakage signature. Check timing, drop it, re-run.
2	**B**	Picking the best of ten on the test set makes the test score optimistic. Validation (or CV) chooses, test is used once.
3	**C**	Selective labels. Non-attenders are *unobserved*, not "not readmitted". Compare populations and follow a random sample (Learn).
4	**B**	K = 1: low bias, high variance, overfit. Choose K on validation.
5	**B**	Training R^2 never falls when inputs are added. Falling test R^2 = overfitting.
6	**B**	Same performance → simpler model: explainable, auditable, less overfit, cheaper to monitor.
7	**B**	Base rate 15% → "call nobody" is already 85% accurate. Ask for the baseline and the error costs.
8	**C**	Evidence must match deployment: recent unseen patients, including groups missing from training.

9 · "A more flexible model always predicts new cases better." (2 p)

False. More flexibility lowers bias but raises variance, so test error is U-shaped: past some point a flexible model fits noise and predicts new cases worse (e.g. K = 1, or Model 3 in Part B with training AUC 0.95 but test AUC 0.66).

10 · "A model with 95% accuracy is always better for the hospital than one with 90%." (2 p)

False. Accuracy counts every error the same, but a missed readmission costs 9,000 and an unnecessary call 1,000; a 90% model that catches more readmissions can have a lower total cost (Part C: 72% accuracy at 0.10 costs 520,000, 88% at 0.50 costs 960,000).

Exam rule for true/false: words like "always", "never", "proves" are almost always false. Give the reason with a concrete counter-example from the scenario.

Part B · Reading and comparing regression models (34 p)

B1 · Probability for a patient over 75 with two prior admissions (3 p)

```
z = −2.20 + 0.69·1 + 0.40·2 = −2.20 + 0.69 + 0.80 = −0.71
e^z = e^−0.71 = 0.492
p = 0.492 / (1 + 0.492) = 0.492 / 1.492 = 0.33
```

About 33% predicted probability of readmission within 30 days.

Write every step. Plug in 1 for the Yes dummy and the count for the per-unit input. Check the answer is between 0 and 1.

B2 · Interpret 0.40; is "0.69 matters more than 0.40" fair? (5 p)

(a) e^0.40 ≈ **1.49**. Each additional prior admission multiplies the odds of readmission by about 1.49 (about 49% higher odds), **holding age_over_75 fixed**. It is a change in odds, not a 49-point rise in probability, and it is an association.

(b) Not fair, because the inputs are on different scales. age_over_75 is a 0/1 switch, while 0.40 is per single admission. Two prior admissions add 0.80 and three add 1.20 to the log-odds, more than 0.69. To compare, use a realistic change for each input (e.g. per standard deviation or over the typical range). And a larger coefficient does not mean a larger causal effect.

B3 · "p < 0.001, so phoning patients with many prior admissions will reduce their readmissions." (6 p)

(a) **Not supported.** The p-value says the *association* between prior admissions and readmission is unlikely to be chance. It says nothing about what a phone call does. Prior admissions mark sicker patients (prediction of risk), but the claim is about an intervention: P(Y | do(call)). High risk is not the same as persuadable. Some high-risk patients will be readmitted whatever we do.

(b) **A/B test:** take eligible discharged patients (e.g. those with ≥ 2 prior admissions) and **randomly assign** each to a phone call or to usual care (no call). Compare the **30-day readmission rate** (and total cost) between the two groups. Random assignment makes the groups alike on everything, measured or not (severity, support at home), so the difference is caused by the call.

(c) Check one: the groups are balanced at baseline; readmissions are recorded for both groups, including at other hospitals; the calls were actually made; the sample is large enough; the outcome was fixed in advance.

B4 · Training data contain only patients discharged home (4 p)

Patients sent to a nursing facility are **0% of the training data** (representativeness / deployment fit). The model has never seen them, so its score for them is an extrapolation: their risk relationships and calibration are unknown, and the score may be badly wrong.

Limit: do not use the model for them; keep the current process (clinician judgment or a fixed call rule). **Learn:** score them in shadow mode and collect their 30-day outcomes, then validate and calibrate before using the model for that group.

B5 · Length of stay = 2.1 + 0.05 × age (4 p)

(a) Among patients aged 60 to 95, each additional **year of age** is associated with **0.05 more days** in hospital on average (about half a day per decade). An association, not a causal effect.

(b) The mean residual rises from −1.1 (60–69) to +1.5 (90+): the line over-predicts for younger and under-predicts for older patients, so the relationship is **curved, not straight** (add age², age bands or a spline). The spread grows from 2.0 to 5.5 days, so predictions are **much less precise for older patients** than the average residual error of 3.4 days suggests. With $R^2 = 0.12$, age explains little of length of stay anyway, and only 40 patients are 90+.

B6 · Model 3: neither coefficient significant (4 p)

(a) **Collinearity.** Age and chronic conditions correlate 0.90, so they carry almost the same information. The model cannot separate their individual contributions, the standard errors inflate (age 0.01 → 0.02, conditions 0.07 → 0.16) and neither is individually significant, even though together they fit as well as before (R^2 0.14). VIF ≈ 1/(1 − 0.81) ≈ 5.3.

(b) **No.** 0.25 ± 2 × 0.16 = −0.07 to 0.57, which includes 0. The estimate is unstable, and in the data age and conditions rarely move separately, so "holding age fixed" is barely observed. It is also only an association.

(c) **Prediction:** collinearity does not hurt predictions much. Keep the model that predicts best on test data (Model 3 or either single one). **Explanation:** do not interpret the separate coefficients. Use one variable or a combined index, report the joint effect, or collect data where the two vary independently.

B7 · Three logistic regressions (4 p)

(a) **Model 3 overfits:** training AUC 0.95 but test AUC 0.66. It is the largest gap and the worst test score, because 40 extra variables let it fit noise (high variance).

(b) **Model 2:** best test AUC (0.76), small train–test gap (0.78 → 0.76) so it generalises, and 8 inputs are still explainable to clinicians. Model 1 (0.70) is the fallback if the extra measures are costly or unavailable.

(c) Check that the 6 clinical measures are **available at discharge** (no leakage) and recorded for all patients, and that the AUC gain actually **lowers cost at the chosen threshold** by more than the data cost.

B8 · Confidence interval for age_over_75 (4 p)

(a) 0.69 ± 2 × 0.25 = **0.19 to 1.19** (odds ratio e^0.19 ≈ 1.2 to e^1.19 ≈ 3.3). Holding prior admissions fixed, patients over 75 plausibly have between about 1.2 and 3.3 times the odds of readmission.

(b) The interval shows the **size and precision** of the effect: it is positive, but it could be small (1.2×) or large (3.3×). The p-value only says the effect is unlikely to be zero.

(c) **No.** Copying rows adds no new information. Standard errors assume independent observations, so duplicates shrink the SE artificially. It is false precision, and the coefficient itself does not change.

Part C · Cost and threshold (29 p)

Test set: 1,000 patients, 150 readmitted, 850 not. TP = flagged and readmitted · FN = not flagged and readmitted · FP = flagged, not readmitted · TN = not flagged, not readmitted.

Threshold	TP	FN	FP	TN	Recall	Precision	Specificity	Accuracy	Cost (9,000 / 1,000)
0.50	45	105	15	835	0.30	0.75	0.98	88.0%	960,000
0.25	85	65	115	735	0.57	0.43	0.86	82.0%	700,000
0.10	120	30	250	600	0.80	0.32	0.71	72.0%	520,000

C1 · Sensitivity and precision at 0.50 (5 p)

```
Sensitivity = TP/(TP+FN) = 45/(45+105) = 45/150 = 0.30
Precision   = TP/(TP+FP) = 45/(45+15)  = 45/60  = 0.75
```

Sensitivity 0.30: the model catches only 30% of the patients who will be readmitted and misses 105 of 150. **Precision 0.75:** 3 out of 4 calls go to patients who really are readmitted, so few calls are wasted.

C2 · Cost at 0.10, choice, accuracy, worth running (7 p)

```
(a) Cost = C_FN·FN + C_FP·FP = 9,000·30 + 1,000·250 = 270,000 + 250,000 = 520,000
```

(b) **The model at 0.10**: 520,000 per 1,000 is the lowest of the four, 330,000 below calling everybody (850,000) and 830,000 below calling nobody. Note that the model at 0.50 (960,000) is worse than simply calling everybody.

(c) Calling nobody is 85% accurate while catching **zero** readmissions, because 85% of patients are not readmitted. Accuracy treats a 9,000 miss like a 1,000 call, so 88% hides that the 0.50 model still misses 105 patients and costs more than calling everyone.

```
(d) Saving vs best baseline = 850,000 − 520,000 = 330,000 per 1,000 patients
    Per year: 330,000 × 4,000/1,000 = 1,320,000
    Net benefit = 1,320,000 − 400,000 = 920,000 per year  (return 3.3× the running
cost)
    Stop if running cost > 1,320,000 per year
```

Worth running. Caveats: assumes the cost estimates and the calibration hold next year, and that the team can make 370 calls per 1,000.

C3 · Break-even; why cost and accuracy move apart (5 p)

```
(a) p* = C_FP / (C_FP + C_FN) = 1,000 / (1,000 + 9,000) = 0.10
```

(b) Accuracy counts errors, cost weighs them. From 0.50 to 0.10 we swap expensive errors for cheap ones: misses fall 105 → 30 (−75 × 9,000 = −675,000) while unnecessary calls rise 15 → 250 (+235 × 1,000 = +235,000). Total errors rise (120 → 280, so accuracy falls to 72%) but cost falls by 440,000. Because readmissions are rare, catching them means more false alarms, which accuracy punishes and cost does not.

C4 · Capacity 200 calls per 1,000 and a cost range of 4,000 to 19,000 (7 p)

(a) **Feasible: 0.50 (60 calls) and 0.25 (200 calls).** 0.10 needs 370 calls, so it is not feasible.

(b) **Choose 0.25.** It is feasible and cheaper than 0.50 at both ends of the range (375,000 vs 435,000 at 4,000; 1,350,000 vs 2,010,000 at 19,000), so it is the best feasible choice whatever the true miss cost. The mistake we can afford is an unnecessary call (1,000); the one we cannot afford is a missed readmission if its true cost is near 19,000.

```
(c) At 19,000: 0.25 → 1,350,000 ; 0.10 → 820,000
    Gain from 370 calls = 1,350,000 − 820,000 = 530,000 per 1,000 patients
    for 170 extra calls → worth up to about 530,000/170 ≈ 3,100 per extra call
    (per year with 4,000 patients: 2,120,000)
```

Worth asking for if extra capacity for 170 calls costs less than about 530,000 per 1,000 patients. But at a miss cost of 4,000 the gain is only 5,000 (375,000 → 370,000), so first ask Finance to narrow the cost range. The value of capacity depends on the true miss cost.

C5 · ROC curve (5 p)

Coordinates (FPR, TPR): 0.50 → (15/850 = 0.02, 0.30); 0.25 → (115/850 = 0.14, 0.57); 0.10 → (250/850 = 0.29, 0.80).

(a) A lower threshold flags more, so its point lies **further up and to the right**. The lowest-left point is 0.50, the middle is 0.25, the highest-right is 0.10 (on the figure, most likely A = 0.50, B = 0.25, C = 0.10; check the coordinates above).

(b) The dashed diagonal is a coin flip (TPR = FPR, AUC 0.50). **AUC 0.83** means that in 83% of pairs, a randomly chosen readmitted patient gets a higher score than a randomly chosen non-readmitted patient.

(c) Choose the **0.10 point (C)**: it matches p* = 0.10 and has the lowest cost (520,000), if capacity allows; with the 200-call limit, B (0.25). The ROC curve shows only the trade-off between TPR and FPR. It contains no costs, no base rate (150 vs 850), no capacity and no calibration, so it cannot pick the best point on its own.

Part D · Audit the AI and decide (25 p)

Item 1 · Five claims (10 p)

Claim	Verdict	Reason
1 · $R^2 = 0.62$ → explains 62% of next year	**Not supported**	That is a training-data fit. Training R^2 is optimistic (it rises with every input) and next year's patients may differ. Only R^2 on new, recent data says how well it predicts.
2 · Pharmacy visits $p < 0.001$ → more visits reduce readmissions	**Not supported**	A p-value shows association, not cause. Patients with more pharmacy visits may differ in health or support (confounding). Only a randomized test can show the effect of adding visits.
3 · 91% accurate → ready	**Not supported**	With a 9% readmission rate, calling nobody is also 91% accurate. We need recall, cost against the baselines, and calibration by group.
4 · At 0.10, flags 120 of the 150 readmitted	**Supported**	Part C table: TP = 120 at 0.10, so recall = 120/150 = 0.80.
5 · Class joiners readmitted less → class cuts readmissions	**Not supported**	Self-selection: patients who chose the class are likely healthier or more motivated. Observational. Randomize the invitation to the class to test it.

Item 2 · Expected vs actual cost by group (7 p)

(a) Trust **new patients** less: actual cost 270,000 is **2.25×** the expected 120,000, while returning patients are close (250,000 vs 230,000, about 9% off). The model under-predicts new patients' risk, most likely because they are thin or missing in the training data.

(b) Check: a **calibration check by group**. Compare predicted and actual readmission rates by score band for new patients on recent outcomes, and count how many new patients were in the training data.

(c) If new patients' actual readmission rate is clearly above their predicted rate (e.g. scored 5% but 15% readmitted), **lower the threshold for new patients** (call more of them) or recalibrate their scores upward; within capacity, keep them on a simple rule (Limit) until recalibrated.

Item 3 · JUDGMENT and ACTION, at most 40 words each (8 p)

JUDGMENT (33 words): The model ranks risk well (test AUC 0.83). Within the 200-call limit, threshold 0.25 costs 700,000 per 1,000 patients, versus 1,350,000 calling nobody. It predicts risk; it does not prove calls prevent readmissions.

ACTION (34 words): From Monday, the discharge-nurse lead calls patients scored ≥0.25 (about 200 per 1,000); a random 10% of them get usual care to measure the calls' effect. Monitor monthly: new patients' actual versus expected readmissions.

What earns the points: numbers that match the tables, the capacity limit respected, a claim and a refusal to over-claim, an owner, and one monitored number.

Appendix · What Assignment 2 adds for the exam

Assignment 2 is built from the same question types. If a question looks like one of these, reuse the reasoning.

Assignment 2 task	Exam twin	The point to make
T1 · Superhost badge coefficient shrinks from Model A (badge alone) to Model B (+ bedrooms, distance, rating)	B2, B3, sign flip	**Confounding.** Badge listings are bigger, closer and better rated, so alone the badge picks up their price premium. With those held fixed the badge effect is smaller. The CI shows the size and precision; the p-value does not.
T1 · Bedrooms coefficient in Model B	B5 (a)	"Holding badge, distance and rating fixed, each extra bedroom is associated with about [b] higher price per night."
T2 · Model C adds max_guests (≈ 2 × bedrooms)	B6	**Collinearity:** bedrooms' SE inflates, max_guests not significant and may have an odd sign, fit unchanged. Predict with either; explain with Model B.
T3 · Model D = Model B + 60 noise columns	A5, B7	Training R^2 rises, test R^2 falls: **overfitting**. Never trust training R^2.
T3 · Test RMSE vs "always predict the mean"	C2 baselines	Always compare with the dumb baseline.
T3 · Predict a 6-bedroom listing (data: 1–4)	B5 (c-type)	**Extrapolation:** outside the data range; the prediction interval is wide and the line may not hold.
T3 · Residuals by distance band	B5 (b)	Systematic pattern by band → price falls with distance in a **curve**, not a line (positive residuals near the centre and far out, negative in between).
T4 · "Will coaching for the badge raise my price?"	B3	Observational association, not an effect. Compare predicted gains from Model B (badge coefficient, +1 bedroom, +0.2 rating × rating coefficient), then **A/B test**: randomly offer coaching to some hosts, compare price changes, check balance and compliance.
T5–T7 · Lab Part B (fraud: costs, capacity, one changed assumption)	C3, C4	p* from costs; feasible thresholds under capacity; cost of the limit; robustness to a cost range; one validity check (calibration for these cases, who is missing, who owns it).
T8 · Odds ratio per SD	B2	"One SD (= __ units) higher [input] multiplies the odds of default by __, holding the others fixed." Model learned only from approved established customers → no evidence for first-time customers; a coefficient is not the effect of changing someone's debt ratio.
T9 · Confusion matrix at 0.50 vs your threshold	C1, C3	Accuracy falls while cost falls: cheap false declines replace expensive missed defaults.
T10 · Expected vs actual cost, first-time vs established	D Item 2	Trust first-time customers less; calibration check by group; lower their threshold or Limit.
T11 · Worth running (20,000 applications, 10,000,000 a year)	C2 (d)	Saving per 1,000 vs best baseline × 20 = saving per year; compare with running cost; stop above that saving.
T12 · JUDGMENT and ACTION blocks	D Item 3	Numbers that match, a refusal to over-claim, an owner, a monitored number.